

# MAT 127A Discussion 9 Notes

Chen Liang

November 25, 2025

## §I More Functional Limits

(Warm-up) Let  $g(x) = \sqrt[5]{x}$ . Prove that  $\lim_{x \rightarrow c} g(x) = \sqrt[5]{c}$  for all point  $c > 0$ . (The identity  $a^5 - b^5 = (a - b)(a^4 + a^3b + a^2b^2 + ab^3 + b^4)$  will be helpful. Try  $a = \sqrt[5]{x}$  and  $b = \sqrt[5]{c}$ .)

*Proof.* Let  $\varepsilon > 0$ . Note that for all  $x \in \mathbb{R}$ , if we set  $a = \sqrt[5]{x}$  and  $b = \sqrt[5]{c}$ , then we have

$$|\sqrt[5]{x} - \sqrt[5]{c}| = |a - b| = \frac{|a^5 - b^5|}{a^4 + a^3b + a^2b^2 + ab^3 + b^4},$$

and if we further assume  $x > 0$ , then we have

$$|\sqrt[5]{x} - \sqrt[5]{c}| \leq \frac{|a^5 - b^5|}{b^4} = \frac{|x - c|}{\sqrt[5]{c^4}}.$$

Now, choose  $\delta = \min\{c, \varepsilon \sqrt[5]{c^4}\}$ . To verify that this choice of  $\delta$  works, fix arbitrary  $x \in \mathbb{R}$  with  $0 < |x - c| < \delta$ , then observe that  $|x - c| < c$  implies  $x > 0$ , and therefore

$$|\sqrt[5]{x} - \sqrt[5]{c}| \leq \frac{|x - c|}{\sqrt[5]{c^4}} < \frac{\varepsilon \sqrt[5]{c^4}}{\sqrt[5]{c^4}} = \varepsilon.$$

■

Recall the sequential criterion for functional limits. Given a function  $f : A \rightarrow \mathbb{R}$  and a limit point  $c$  of  $A$ , the following two statements are equivalent:

- (i)  $\lim_{x \rightarrow c} f(x) = L$ .
- (ii) For all sequences  $(x_n)$  in  $A$  satisfying  $x_n \neq c$  for all  $n$  and  $(x_n) \rightarrow c$ , it follows that  $(f(x_n)) \rightarrow L$ .

Use the algebraic limit theorem for sequential limits to prove the following. Let  $f$  and  $g$  be functions defined on a domain  $A \subseteq \mathbb{R}$ , and assume  $\lim_{x \rightarrow c} f(x) = L$  and  $\lim_{x \rightarrow c} g(x) = M$  for some limit point  $c$  of  $A$ . Then,  $\lim_{x \rightarrow c} f(x)g(x) = LM$ .

*Proof.* It suffices to show that for all sequences  $(x_n)$  in  $A$  satisfying  $x_n \neq c$  for all  $n$  and  $(x_n) \rightarrow c$ , it follows that  $(f(x_n)g(x_n)) \rightarrow LM$ . Since we have  $\lim_{x \rightarrow c} f(x) = L$  and  $\lim_{x \rightarrow c} g(x) = M$ , the sequential criterion for functional limits tells us that  $(f(x_n)) \rightarrow L$  and  $(g(x_n)) \rightarrow M$ . Thus, by the algebraic limit theorem for sequential limits, we have  $(f(x_n)g(x_n)) \rightarrow LM$ , as desired. ■

This is one of pieces of the algebraic limit theorem for functional limits. The full algebraic limit theorem for functional limits says that under the same assumptions, we have

- (i)  $\lim_{x \rightarrow c} kf(x) = kL$  for all  $k \in \mathbb{R}$ ,
- (ii)  $\lim_{x \rightarrow c} (f(x) + g(x)) = L + M$ ,
- (iii)  $\lim_{x \rightarrow c} f(x)g(x) = LM$ ,
- (iv)  $\lim_{x \rightarrow c} f(x)/g(x) = L/M$  provided  $M \neq 0$ .

Using the algebraic limit theorem, one can easily show that polynomials are continuous.

Function composition may not behave nicely with functional limits.

(Exercise 4.3.4) Assume  $f$  and  $g$  are defined on all of  $\mathbb{R}$  and that  $\lim_{x \rightarrow p} f(x) = q$  and  $\lim_{x \rightarrow q} g(x) = r$ .

1. Disprove the claim that  $\lim_{x \rightarrow p} g(f(x)) = r$  by verifying that

$$f(x) = 0 \quad \text{and} \quad g(x) = \begin{cases} 1 & \text{if } x = 0, \\ 0 & \text{if } x \neq 0 \end{cases}$$

is a counterexample.

2. Show that if  $g(q) = r$ , then  $\lim_{x \rightarrow p} g(f(x)) = r$ .

*Proof.*

1. We have  $\lim_{x \rightarrow 0} f(x) = 0$  and  $\lim_{x \rightarrow 0} g(x) = 0$ , but  $\lim_{x \rightarrow 0} g(f(x)) = \lim_{x \rightarrow 0} 1 = 1$ .
2. Let  $\varepsilon > 0$ . Since  $\lim_{x \rightarrow q} g(x) = r$ , there exists  $\delta' > 0$  such that if  $0 < |y - q| < \delta'$ , then  $|g(y) - r| < \varepsilon$ . Moreover, since  $g(q) = r$ , we have  $|g(y) - r| = 0 < \varepsilon$  for  $y = q$ , and so we can conclude that if  $|y - q| < \delta'$ , then  $|g(y) - r| < \varepsilon$ . Now, we can find the desired  $\delta > 0$  by noting that  $\lim_{x \rightarrow p} f(x) = q$  implies that there exists  $\delta > 0$  such that if  $0 < |x - p| < \delta$ , then  $|f(x) - q| < \delta'$ , which then implies  $|g(f(x)) - r| < \varepsilon$ .

■

## §II Continuous Functions

Recall the characterization of continuity. Let  $f : A \rightarrow \mathbb{R}$ , and let  $c \in A$ . The function  $f$  is continuous at  $c$  if and only if any one of the following three conditions is met:

- (i) (Definition) For all  $\varepsilon > 0$ , there exists a  $\delta > 0$  such that  $|x - c| < \delta$  (and  $x \in A$ ) implies  $|f(x) - f(c)| < \varepsilon$ ;
- (ii) For all  $\varepsilon > 0$ , there exists a  $\delta > 0$  with the property that  $f(V_\delta(c) \cap A) \subseteq V_\varepsilon(f(c))$ ;
- (iii) For all  $(x_n) \rightarrow c$  (with  $x_n \in A$  for all  $n$ ), it follows that  $(f(x_n)) \rightarrow f(c)$ .

If  $c$  is a limit point of  $A$ , then the above conditions are equivalent to

- (iv)  $\lim_{x \rightarrow c} f(x) = f(c)$ .

Note that if  $c$  is an isolated point of  $A$ , it still makes sense to say that  $f$  is continuous at  $c$  even though it does not make sense to talk about  $\lim_{x \rightarrow c} f(x)$ .

The previous exercise tells us that composition of continuous functions is continuous, and so composition of **continuous** functions behaves nicely with functional limits.

Let  $F$  be a closed subset of  $\mathbb{R}$  and  $f$  be a continuous function defined on  $\mathbb{R}$ . Decide if the following statements are always true. If not, provide a counterexample.

1. The set  $f(F)$  is closed.
2. The set  $f^{-1}(F)$  is closed.

*Proof.*

1. This is not always true. Consider  $f(x) = e^x$  and  $F = \mathbb{R}$ , then we have  $f(F) = (0, \infty)$ , which is not closed.

**Remark:** Note the the only examples are those with unbounded  $F$  because image of a compact set under a continuous function is always compact.

2. This is always true. To see this, let  $(x_n)$  be a sequence in  $f^{-1}(F)$  converging to some  $x$ . We wish to show that  $x \in f^{-1}(F)$ , which is to say  $f(x) \in F$ . Since  $f$  is continuous at  $x$ , we have  $(f(x_n)) \rightarrow f(x)$ , and because  $f(x_n) \in F$  for all  $n$  and  $F$  is closed, we must have  $f(x) \in F$ , as desired.

■